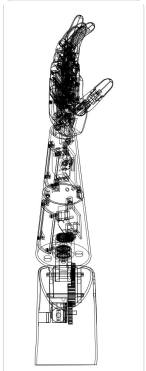
INPUT **Two views of one part?**

2D drawing

OpenCascade



CAD render

Blender



PROMPT **One binary question**

SYSTEM

You are an expert mechanical engineer performing quality inspection.

USER

<image 1> <image 2> The first image is a 2D engineering drawing. The second is a render of a 3D CAD model. Do they show the same part? Answer with exactly one word: **Yes** or **No**.

MODEL **Small open-weight VLM**

Qwen3-VL
2B parameters

Gemma-3
4B parameters

LoRA · ours

vision encoder
image → patch tokens

LoRA

language model
tokens → Yes / No

zero-shot	all weights frozen
text	LoRA on the language model
text + vision · ours	**LoRA on both**

ANSWER **Yes / No**

read from the log-prob of Yes vs No



Yes

same part